
From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

1 Even untrained random embeddings, when injected into an LLM input, change
2 the model’s reasoning behavior in non-trivial ways. Recent *soft prompt* methods
3 optimize trainable virtual-token embeddings to improve mathematical reasoning,
4 but it is unclear whether the gain comes from the *learning* or from the *injection*
5 itself. To separate the two, we strip the training step entirely: a Random Soft
6 Prompt (RSP) is a continuous vector sampled from an isotropic Gaussian fitted to
7 the entrywise mean and variance of the pretrained embedding table, and injected
8 as is. Two simple observations carry the analysis. First, the influence of an
9 RSP is largest in the early generation steps and naturally fades as more tokens
10 are appended—each layer’s attention assigns less weight to the random tokens
11 once the autoregressive cache has filled in. Second, a single drawn RSP is fixed
12 within one rollout and selects one branch different from the baseline; because each
13 rollout draws a fresh RSP, branches from different rollouts accumulate, and a token
14 rarely seen by the baseline can enter the candidate set in some rollouts. We make
15 both observations quantitative inside the transformer, and verify them with three
16 experiments: baseline-accuracy recovery, an early-token diversification followed
17 by late-stage stabilization, and Pass@ N saturation that disappears once the same
18 RSP is shared across samples. We close by discussing the implications of this
19 diversity for GRPO training, where trajectory diversity is critical for learning signal
20 quality.

21 1 Introduction

22 As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increas-
23 ingly expensive. The dominant alternative is parameter-efficient adaptation, in which only a small
24 set of additional parameters is introduced [Hu et al., 2022, Houlsby et al., 2019]. Among these,
25 *soft prompts* insert learnable continuous embeddings outside the vocabulary into the input, steering
26 model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. The
27 same paradigm has been broadly adopted to enhance mathematical reasoning in recent work [Kang
28 et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026], where the com-
29 mon structure is to *optimize* continuous embeddings outside the vocabulary to improve downstream
30 accuracy.

31 There is, however, a clue that the effect does not come from optimization alone: injecting untrained
32 random continuous embeddings already shifts the output distribution. For example, applying a chat
33 template to a base model that was not fine-tuned for chat causes a format mismatch that degrades
34 reasoning accuracy, but appending random embeddings recovers a substantial portion of that loss. If
35 simple noise were merely perturbing the model, accuracy should worsen instead. This suggests that
36 the effect arises not from *learned content* but from some property of the *injection itself*—a question
37 prior evaluations, focused on end-task accuracy, have not separated from learned content.

We investigate Random Soft Prompts (RSPs), continuous vectors injected without any training. RSP draws each entry from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table. The role of randomness is not to imitate the embedding distribution; it is to ensure that the same prompt produces *independent* perturbations across rollouts, since exploration is what requires this independence. The point is not the number of trials, but that each trial opens early-token branches differently. We answer two questions: why does a single RSP injection only stir the *early* tokens and then fade, and why do reasoning paths accumulate across trials when fresh RSPs are drawn? The first question is answered by a token-wise self-regulation that the transformer’s attention decomposition naturally produces; the second is answered by an exponential accumulation chain in which N independent retries quickly saturate small per-trial entry probabilities. A single RSP fixes one branch within its rollout; as trials accumulate, baseline-excluded candidates enter the candidate set in some trials, and the events accumulate as a distribution. Finally, we discuss the implications of this diversity for GRPO [Shao et al., 2024] training.

Our contributions are as follows.

- We empirically analyze how RSPs affect LLM generation: RSPs raise early-token entropy and stabilize back to baseline at later steps, and combined with temperature sampling produce more diverse reasoning trajectories.
- We explain why soft prompts work without training at the transformer level, with both natural-language intuition and quantitative bounds. The entire effect of RSP is controlled by a single attention-side scalar that automatically weakens as the cache grows; an isotropic distribution places no prior weight on any direction while assigning positive probability to every direction.
- We discuss what RSP-induced diversity implies for GRPO training. RSP is a *rank-changing* perturbation while temperature is *rank-preserving*—this difference makes the two methods provide orthogonal exploration gains.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model’s own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model’s reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024]

adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time. At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity. In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters. RSPs differ from these approaches in the following respects. (1) RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. (2) RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. (3) RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Random Soft Prompts and Why They Work

This section defines Random Soft Prompts (RSPs) and explains their behavior in two parts. A single RSP draw is fixed within one rollout and selects one branch different from baseline. Its influence is meaningfully large at early tokens (§3.2) and automatically converges to zero as decoding proceeds (§3.3). Because each rollout draws a fresh RSP, the across-rollout differences accumulate into Pass@ N improvements at an exponential rate (§3.4). The body keeps theorem statements and intuition; all formal proofs are deferred to Appendix E.

3.1 Random Soft Prompt

The construction is simple: borrow the entrywise mean and scale of the pretrained token embedding table to draw a continuous vector of the same shape, and concatenate it before, inside, or after the prompt.

Formally, let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the token embedding matrix (V is the vocabulary size, d the hidden dimension), and write its entrywise mean and standard deviation as $\mu_E := \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E := \text{std}(\mathbf{W}_E)$. An RSP of length L is a sequence of continuous vectors $\mathbf{H} = [h_1, \dots, h_L] \in \mathbb{R}^{L \times d}$ drawn independently from

$$h_j \sim \mathcal{N}(\mu_E \mathbf{1}_d, \sigma_E^2 \mathbf{I}_d), \quad j = 1, \dots, L. \quad (1)$$

The centered form $\bar{h}_j := h_j - \mu_E \mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is a zero-mean isotropic Gaussian. The injection position into the input embedding sequence $\mathbf{X}$ is one of three: prefix $[\mathbf{H}; \mathbf{X}]$, infix (insert $\mathbf{H}$ inside $\mathbf{X}$), or suffix $[\mathbf{X}; \mathbf{H}]$ [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]. For repeated-sampling protocols, the prompt draws and decoding randomness across compared trials are assumed independent.

The role of randomness is not to imitate the embedding distribution. It is to ensure that the same prompt produces *independent* perturbations across rollouts—and exploration is precisely what requires this independence.

3.2 Early exploration: a lower envelope

RSP does not add to the hidden state directly; it enters only through attention weights. Hence the entire instantaneous influence of RSP on one layer collects into a single scalar—the attention mass assigned to RSP tokens. Consider self-attention layer ℓ with n real tokens and L RSP tokens, attention weights $\alpha_{ij}^{(\ell)}$, and value vectors $\mathbf{v}_{j,i}^{(\ell)}, \mathbf{v}_{r,j}^{(\ell)}$ on the real and RSP sides. Define $w_{r,i}^{(\ell)} := \sum_{j'=1}^L \alpha_{i,n+j'}^{(\ell)}$.

133 The attention output decomposes exactly as

$$\mathbf{x}_i^{(\ell+1)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{x}}_i^{(\ell+1)} + \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{x}}_i^{(\ell+1)} := \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j'=1}^L \alpha_{i,n+j'}^{(\ell)} \mathbf{v}_{r,j'}^{(\ell)}. \quad (2)$$

134 This identity uses only softmax normalization (derivation: Appendix E). Since $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq$
 135 $w_{r,i}^{(\ell)} \max_{j'} \|\mathbf{v}_{r,j'}^{(\ell)}\|$, the same scalar $w_{r,i}^{(\ell)} \in [0, 1]$ controls both the attenuation ratio $1 - w_{r,i}^{(\ell)}$ on
 136 the real-token signal and the upper bound on the random contribution’s magnitude. The behavior of
 137 RSP reduces to tracking one value rather than a multidimensional perturbation.

138 The key question is whether this value is *meaningfully nonzero* at early decoding. The following
 139 lower-envelope statement provides this guarantee. Let $\mathbf{q}_i^{(\ell)}$ be the query and $\mathbf{k}_j^{(\ell)}, \mathbf{k}_{r,j'}^{(\ell)}$ the real and
 140 RSP keys at layer ℓ .

141 **Proposition 1** (Lower envelope for early exploration). *Define the reverse logit gap $\Delta_i'^{(\ell)} :=$*
 142 *$\max_{j \in [n]} (\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_j^{(\ell)} - \min_{j' \in [L]} (\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_{r,j'}^{(\ell)}$. Then in scaled dot-product attention, $w_{r,i}^{(\ell)} \geq$*
 143 *$L/(L + n \exp(\Delta_i'^{(\ell)}/\sqrt{d}))$.*

144 The proof is in Appendix E (symmetric to Theorem 1 below). The bound equals 1 at $n = 0$ and stays
 145 close to 1 while n is small, provided $\Delta_i'^{(\ell)}$ is not excessively large. Hence *when the reverse gap does*
 146 *not blow up*, early decoding inherits a strictly positive lower bound on RSP influence—a single very
 147 poor RSP key score, however, can inflate $\Delta_i'^{(\ell)}$ and weaken the bound.

148 3.3 Late convergence: an upper envelope and annealing

149 The symmetric question is whether the same scalar *converges to zero* at late decoding. Define the
 150 forward logit gap $\Delta_i^{(\ell)} := \min_{j \in [n]} (\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_j^{(\ell)} - \max_{j' \in [L]} (\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_{r,j'}^{(\ell)}$ ($\Delta_i^{(\ell)} \leq \Delta_i'^{(\ell)}$).

151 **Theorem 1** (Upper envelope under annealing). *In scaled dot-product attention, $w_{r,i}^{(\ell)} \leq L/(L +$*
 152 *$n \exp(\Delta_i^{(\ell)}/\sqrt{d}))$. The right-hand side is strictly decreasing in n at fixed L and $\Delta_i^{(\ell)}$, and converges*
 153 *to zero as $n \rightarrow \infty$ in the fixed-gap regime.*

154 The proof is in Appendix E. Combined with Proposition 1, we obtain a two-sided sandwich:

$$\frac{L}{L + n \exp(\Delta_i'^{(\ell)}/\sqrt{d})} \leq w_{r,i}^{(\ell)} \leq \frac{L}{L + n \exp(\Delta_i^{(\ell)}/\sqrt{d})}. \quad (3)$$

155 Both envelopes equal 1 at $n = 0$ and tend to 0 as $n \rightarrow \infty$, but *at different rates* ($\Delta_i^{(\ell)} \leq \Delta_i'^{(\ell)}$).
 156 Early on, both envelopes hover near 1; late on, both vanish. The theorem does not assert step-by-step
 157 monotonic decay of the realized mass—queries, keys, and hidden states co-evolve so both gaps shift
 158 over time. What it does guarantee is that the *fixed-gap envelope narrows under cache growth alone*.

159 This means RSP behaves at each attention site as a local perturbation that briefly acts and then weakens.
 160 Defining the disturbance energy $V_{\eta,i}^{(\ell)} := \mathbb{E} \|\boldsymbol{\eta}_i^{(\ell)}\|^2$, a finite second moment $M_v^{(\ell)} := \max_{j'} \mathbb{E} \|\mathbf{v}_{r,j'}^{(\ell)}\|^2$
 161 yields $V_{\eta,i}^{(\ell)} \leq M_v^{(\ell)} (w_{r,i}^{(\ell)})^2$ (Appendix E.1), and combining with Theorem 1 gives $V_{\eta,i}^{(\ell)} = O(1/n^2)$
 162 in the fixed-gap regime. This is an upper bound at *a single attention site*; it does *not* claim that the full
 163 hidden-state trajectory $\|h_t^{\text{RSP}} - h_t^{\text{base}}\|$ decreases or that the transformer minimizes a global energy.

164 3.4 Branch opening and Pass@ N accumulation

165 The two envelopes outline the range within which RSP can influence a single rollout. We now ask
 166 how that influence appears in the output distribution and how it accumulates across rollouts.

167 We first justify the distribution choice. Theorem 1 and Proposition 1 use only attention identities
 168 and logit gaps—neither relies on isotropy. Branch opening, however, requires two distributional
 169 properties: an *isotropic covariance* that distributes the variance budget uniformly over directions, and
 170 *full support* that places positive probability on every nonempty open subset of the surrogate. The
 171 next theorem formalizes the first property at the per-RSP-token level—i.e., for the distribution of a
 172 single $\bar{h} \in \mathbb{R}^d$.

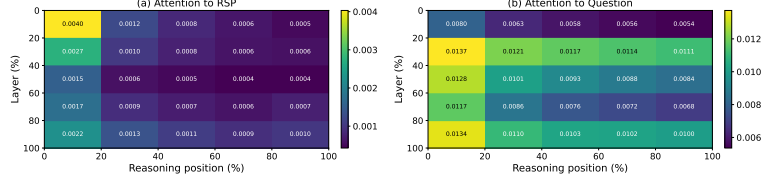


Figure 1: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). (a) RSP-token attention decreases along the reasoning-position axis (horizontal), consistent with the two-sided envelope of Eq. (3). The additional decrease along the layer axis (vertical) is a separate empirical observation suggesting that deeper layers sharpen the real-vs-random logit gap; it does not follow from Theorem 1 alone. (b) Question-token attention remains comparatively uniform. The reported quantity is the per-token mass of Appendix F, equal to $w_{r,i}^{(\ell)}/L$ averaged over heads and samples.

Theorem 2 (Direction-agnostic minimax coverage). *Let D be any zero-mean distribution on $\mathbb{R}^d$ with covariance Σ_D and trace budget $\text{tr}(\Sigma_D) \leq \rho^2$. Then $\min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h) = \lambda_{\min}(\Sigma_D) \leq \rho^2/d$, with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.*

The proof is in Appendix E. Equality holds for *any* distribution with isotropic covariance—Gaussianity is not the unique answer. Fixed prompts ($\Sigma_D = 0$), PCA-aligned noise, vocabulary-token sampling, and embedding-covariance matching are all anisotropic and therefore strictly suboptimal in at least one direction. The reason to pick the Gaussian is the second property: an isotropic Gaussian has positive density everywhere on $\mathbb{R}^d$, so it commits to no particular direction (Lemma 1). The two properties together enable the branch-opening guarantee below.

We now formalize branch opening on the surrogate. Linearizing the logits around $\bar{h} = 0$, define the affine surrogate $z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}$ with $a_i := \nabla_{\bar{h}} z_i(0)$. For a baseline-excluded token i , the top- K entry region is

$$\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}.$$

This matches the actual top- K membership condition.

Proposition 2 (Conditional top- K entry). *Suppose $\mathcal{G}_{i,K}$ contains a nonempty open set. Since the isotropic Gaussian’s full support assigns positive probability to it, $\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$.*

The proof is in Appendix E. The mechanism guarantees that *branches open* ($\mu_i > 0$); whether such a region exists is a task-side condition on the surrogate that is not automatically guaranteed. To lift this single-draw probability to task accuracy, we add two task-side assumptions: (M1) each independent RSP run produces a trajectory in the correct trajectory set with marginal probability at least $p_{\min}(x) > 0$, and (M2) the N runs are mutually independent.

Proposition 3 (Multi-path Pass@ N). *Under (M1)–(M2), with N independent RSP runs,*

$$\Pr(\text{Pass@}N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N \geq 1 - \exp(-N p_{\min}(x)).$$

The proof is in Appendix E. (M1) is task-side and not derivable from the mechanism alone—the mechanism is responsible only for *admitting* baseline-excluded tokens into the candidate set, while whether an admitted branch is *productive* depends on the task. Sharing one RSP across all samples eliminates this accumulation, so *independence* is essential. The empirical signature of (M1) is reported in §4.

4 Empirical Validation

We now examine how the explanation in §3 appears in actual LLM generation through three scenes. First, we ask whether RSP preserves baseline accuracy without breaking it, and recovers it in some settings (§4.2). Second, we check whether the early diversification followed by late stabilization—predicted by the two-sided envelope—is observed empirically (§4.3). Finally, we contrast Pass@ N between sharing one RSP across samples versus drawing a fresh RSP per sample, directly probing whether *independence* is the key (§4.4).

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

Model	MATH-500		GSM8K		AIME24	
	Baseline	RSP	Baseline	RSP	Baseline	RSP
<i>Instruct Models</i>						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
<i>Base Models (Raw Text)</i>						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
<i>Base Models + ChatML (Format Mismatch)</i>						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	84.20	83.80	82.80	83.00
	GSM8K	<u>95.45</u>	95.68	<u>95.30</u>	95.00	94.84
	AIME24	13.33	<u>16.67</u>	23.30	<u>16.67</u>	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	<u>75.20</u>	<u>75.20</u>	76.00	72.40
	GSM8K	85.29	85.75	<u>85.70</u>	84.46	84.53
	AIME24	<u>10.00</u>	20.00	6.70	6.67	<u>10.00</u>

4.1 Experimental Setup

We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates—seven configurations in total. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity, and the answer-extraction pipeline uses fallbacks to score only the final answer. RSPs follow the definition of §3.1 and are concatenated at one of three positions (prefix, infix, suffix) with a freshly drawn **H** for each batch. Full experimental details and per-position results are in Appendix A.

4.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model’s reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to +29 pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method’s distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

4.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model’s output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 2 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

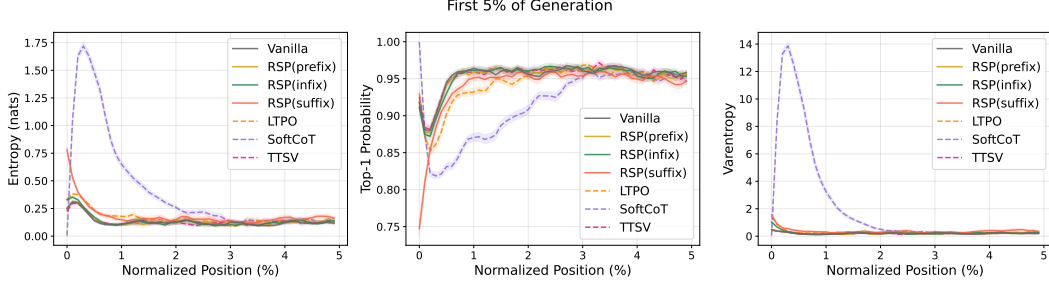


Figure 2: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

4.4 Does RSP Induce Trajectory Diversity?

Section 4.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP’s first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. We test whether RSP modifies the first-token distribution in a qualitatively different way from temperature scaling, which preserves token rankings while only flattening probabilities. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at $\tau=1.0$ as reference, we compare RSP at $\tau=1.0$ against the baseline at $\tau=2.0$ (the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline’s top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix B).

Table 3: First-token distribution metrics measured against the baseline at $\tau=1.0$, comparing the baseline at $\tau=2.0$ with RSP at $\tau=1.0$.

Metric	$\tau=2.0$	RSP
Spearman ρ	1.0	0.651
Mass outside top-10	5.15%	27.64%
JS divergence	0.086	0.231

Table 3 reports the contrast. Even at $\tau=2.0$, only 5.15% of probability mass falls outside the baseline’s top-10. RSP at $\tau=1.0$ reaches 27.64%. The Spearman rank correlation tells the story: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, ρ relative to the baseline is exactly 1. RSP, by contrast, reranks the token distribution itself, reducing ρ to 0.651—an effect temperature cannot produce at any τ . The Jensen–Shannon divergence (0.231) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by about $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by about $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Outcome-level: Pass@ n scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@ n [Chen et al., 2021], the probability that at least one of n sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: (i) *Baseline*, $\tau=1.0$: temperature sampling only; (ii) *RSP (fixed seed)*, $\tau=1.0$: a single RSP shared across all samples; (iii) *RSP (indep. seed), greedy*: a different RSP per sample without temperature; and (iv) *RSP (indep. seed)*, $\tau=1.0$: a different RSP per sample combined with temperature sampling.

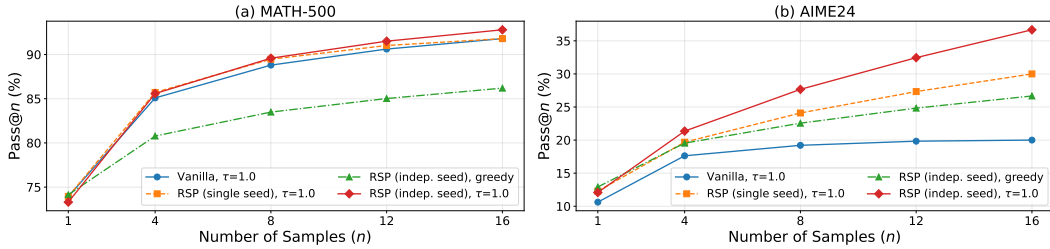


Figure 3: Pass@ n scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition (iv) achieves the highest Pass@ n on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $n = 16$. Condition (ii) shows no improvement over the baseline, suggesting that the diversity gain likely depends on varying the random embeddings across samples rather than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, (iv) achieves higher Pass@1 than the baseline. Together these

307 observations suggest that combining independent RSPs with temperature sampling may yield greater
308 trajectory diversity while preserving single-sample accuracy.

309 5 Conclusion

310 We analyzed Random Soft Prompts (RSPs), continuous vectors drawn from an isotropic Gaussian
311 fitted to the entrywise statistics of the pretrained embedding table and injected without any training.
312 The analysis proceeded through a lower envelope for early exploration (§3.2), an upper envelope and
313 annealing for late convergence (§3.3), and the combination of isotropy, surrogate-level branch opening,
314 and independent resampling that closes with the exponential saturation lower bound $\Pr(\text{Pass}@N) \geq$
315 $1 - \exp(-N p_{\min})$ (§3.4). The three experiments of §4 confirmed the predictions of this inequality:
316 baseline accuracy recovery, the auto-annealing pattern, and the disappearance of accumulation when
317 sharing a single RSP. Because RSP is rank-changing while temperature is rank-preserving, the two
318 are complementary, suggesting a sampling-stage remedy for the entropy collapse in GRPO [Shao
319 et al., 2024] beyond DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026]. Training-time validation
320 against these baselines, extension beyond mathematical reasoning, and a principled choice of RSP
321 length and injection position remain open.

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516 A Experimental Setup and Full Accuracy Breakdown

517 Table 5 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to
518 the best-across-positions results summarized in Table 1 in the main text. Table 6 lists the number of
519 RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are
520 conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation
521 length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is
522 fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-
523 Math-1.5B variants).

Table 5: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
<i>Instruct Models</i>				
LLaMA-3.1-8B-Instruct	Baseline	52.20	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	10.00 (+3.3)
	Suffix	<u>49.40</u> (−2.8)	86.73 (+1.0)	10.00 (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	84.20 (+1.0)	95.60 (+0.2)	16.67 (+3.3)
	Suffix	<u>83.40</u> (+0.2)	95.68 (+0.2)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	<u>13.33</u> (+3.3)
	Infix	75.20 (+2.2)	85.75 (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>				
Qwen2.5-Math-7B	Baseline	72.20	84.15	6.67
	Prefix	<u>71.60</u> (−0.6)	84.31 (+0.2)	16.67 (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	16.67 (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	77.86	16.67
	Prefix	64.40 (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>				
Qwen2.5-Math-7B	Baseline	52.20	58.30	23.33
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	<u>62.00</u> (+9.8)	69.07 (+10.8)	23.33 (+0.0)
	Suffix	70.40 (+18.2)	75.97 (+17.7)	23.33 (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	63.40 (+29.0)	67.02 (+29.6)	20.00 (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 6: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

524 A.1 RSP Injection Positions and Prompt Templates

525 Below we show the prompt structure for each injection position across all model types. Blue text
526 indicates RSP embeddings, and purple text indicates special tokens.

527 A.1.1 Prefix

528 RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

529 ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

531 LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

533 Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{>.
```

535 A.1.2 Infix

536 A description text and special tokens are inserted into the prompt. The special token positions are
537 then replaced with RSP embeddings at the embedding level.

538 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{>
<|im_start|>user
{question}

There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

540 LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{>
<|start_header_id|>user<|end_header_id|>
```

541

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens:

```
<reserved_special_token_0>...  
<reserved_special_token_L>  
<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

542

543 Raw Text (Qwen Base).

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens: $\langle \text{special_token_1} \rangle \dots \langle \text{special_token_L} \rangle$

Please reason step by step, and put your final answer within `\boxed{\}`.

544

545 A.1.3 Suffix

546 For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token.

547 For raw text models, RSP embeddings are concatenated at the end of the prompt.

548 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system  
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>  
<|im_start|>user  
{question}[Random Embeddings (L tokens)]<|im_end|>  
<|im_start|>assistant
```

549

550 LLaMA Chat Template.

```
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023  
Today Date: 26 Jul 2024  
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>  
<|start_header_id|>user<|end_header_id|>  
{question}[Random Embeddings (L tokens)]<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

551

552 Raw Text (Qwen Base).

```
{question}  
Please reason step by step, and put your final answer within \boxed{}. [Random  
Embeddings (L tokens)]
```

553

554 A.2 Answer Extraction and Grading

555 The final answer is extracted from the model output through a fallback chain. Each step is attempted
556 in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. "final answer is \$...\$. I hope" pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following "the answer is"
4. Text following "final answer is"
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing \$, \left, \right, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose, rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature = 0) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 7 reports the per-benchmark hyperparameters.

Table 7: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 4 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 8 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 8: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle-last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 2 (main text) and Figure 4.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

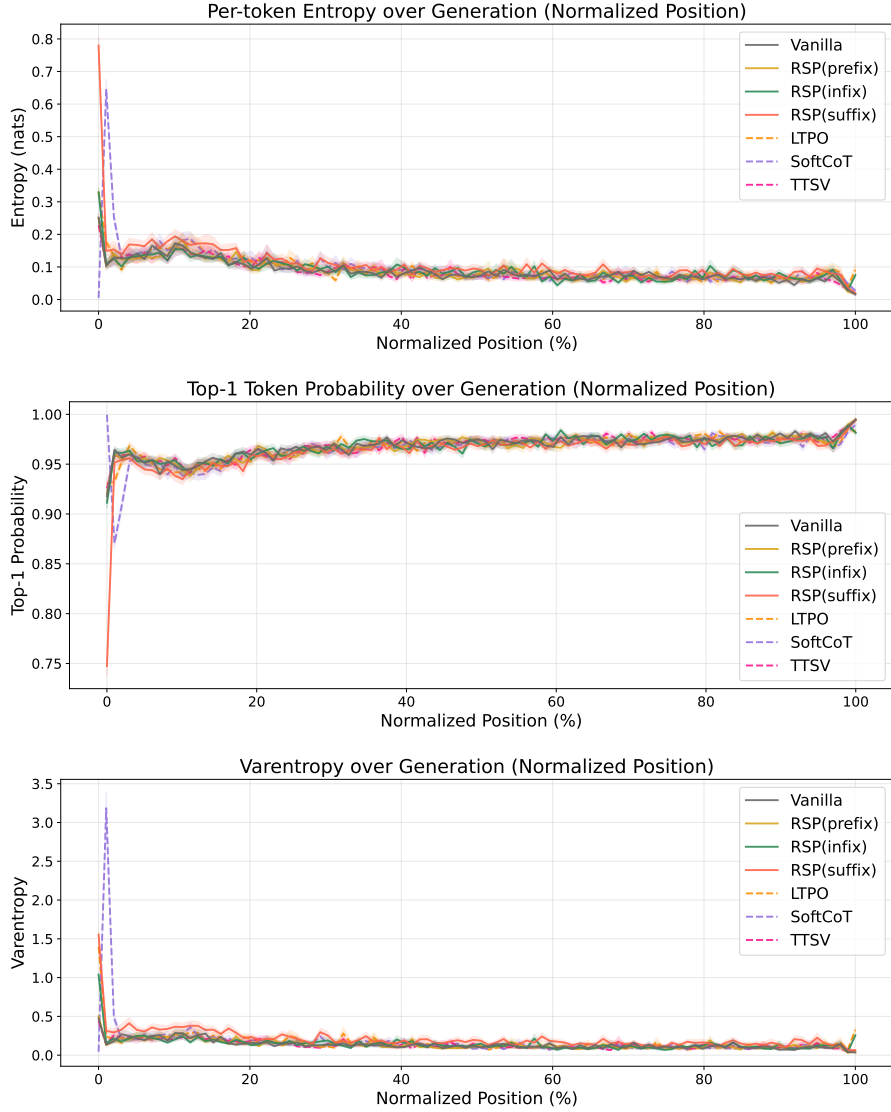


Figure 4: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

580 B Token-Level Distribution Metrics

581 This appendix provides formal definitions for the three metrics used in the Token-level analysis of
 582 Section 4.4. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate
 583 distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are
 584 computed analytically from saved logits at the first generation step.

585 **Spearman rank correlation.** Spearman’s ρ is the Pearson correlation coefficient between the token
 586 ranks of two distributions:

$$\rho = \text{Pearson}(\text{rank}(P_{\text{target}}), \text{rank}(P_{\text{base}})). \quad (4)$$

587 Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens
 588 is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore
 589 indicates rank reorganization beyond what temperature scaling can produce.

590 **Mass outside top- K .** Let $\text{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under
 591 P_{base} . The probability mass that the candidate distribution places *outside* this set is

$$\text{MassOutside}_K = 1 - \sum_{t \in \text{TopK}(P_{\text{base}})} P_{\text{target}}(t). \quad (5)$$

592 This directly measures support expansion: how much probability the candidate assigns to tokens that
 593 are not in the baseline’s preferred set. Reported values use $K = 10$.

594 **Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q). \quad (6)$$

595 We use base-2 logarithms so that $\text{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a
 596 sentinel logit (-10^9) before softmax, which is negligible for our setting since the top-5 covers more
 597 than 99.9% of the probability mass.

C Why Random Prompts Are Semantically Neutral and Direction-Agnostic

This appendix formalizes the “why random” claim used in Section 3.4. The point is not that random prompts contain hidden task information. The point is structural: if the perturbation is meant to act as a generic exploration device rather than as a learned prompt prior, then it should be resampled across runs, centered so that it does not induce a fixed first-order drift, and direction-agnostic so that it does not under-serve whichever local branch direction matters for the current prompt.

C.1 Semantic Neutrality

Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h}/\|\bar{h}\|$ whenever $\bar{h} \neq 0$. Let $u_1, \dots, u_M \in \mathbb{S}^{d-1}$ be any fixed unit vectors representing semantic anchor directions.

Lemma 1 (Semantic neutrality). *For any $\delta \in (0, 1)$,*

$$\Pr \left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2 \log(2M/\delta)}{d-1}} \right] \geq 1 - \delta.$$

Hence the normalized direction of an isotropic Gaussian prompt is simultaneously nearly orthogonal to every fixed finite collection of semantic anchors with high probability.

Proof. Because $\bar{h}$ is isotropic Gaussian, its normalized direction $\hat{h}$ is uniformly distributed on the sphere $\mathbb{S}^{d-1}$. For a fixed unit vector $u \in \mathbb{S}^{d-1}$, the marginal $\hat{h}^\top u$ satisfies the standard spherical concentration bound [Vershynin, 2018, Theorem 3.4.6][Boucheron et al., 2013, Section 3.2],

$$\Pr(|\hat{h}^\top u| \geq t) \leq 2 \exp\left(-\frac{(d-1)t^2}{2}\right) \quad \text{for } t \in (0, 1),$$

which follows from the spherical-cap area bound $\sigma\{\hat{h} \in \mathbb{S}^{d-1} : \hat{h}^\top u \geq t\} \leq \frac{1}{2}(1-t^2)^{(d-1)/2}$ together with $(1-t^2)^{(d-1)/2} \leq \exp(-(d-1)t^2/2)$ derived from $\log(1-x) \leq -x$ for $x \in [0, 1)$. Applying this bound to each u_r and using a union bound,

$$\Pr \left[\max_{r \in [M]} |\hat{h}^\top u_r| \geq t \right] \leq 2M \exp\left(-\frac{(d-1)t^2}{2}\right).$$

Setting

$$t = \sqrt{\frac{2 \log(2M/\delta)}{d-1}}$$

gives the claim. If the displayed value of t exceeds 1, the statement is trivial because $|\hat{h}^\top u_r| \leq 1$ always. $\square$

The contrast with a token embedding is immediate. If $e_s \neq 0$ is a fixed token embedding and $\hat{e}_s := e_s/\|e_s\|$, then choosing the anchor direction $u = \hat{e}_s$ gives

$$|\hat{e}_s^\top u| = 1.$$

A token prompt is therefore maximally aligned with at least one semantic direction, whereas an isotropic random prompt is nearly orthogonal to every fixed finite set of such directions. This is the sense in which tokens inject semantic commitment while random prompts act as semantically uncommitted perturbations.

C.2 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j , and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^\top \bar{h}.$$

Proposition 4 (No systematic first-order drift). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij}$$

for every token pair (i, j) . If a deterministic prompt h_0 is reused across runs, then

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^\top h_0$$

is a fixed directional bias.

630 *Proof.* The centered case is immediate from linearity of expectation:

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

631 The deterministic case follows by substitution. $\square$

632 The proposition does not say that a centered perturbation leaves every *realization* unchanged; it says
 633 there is no systematic first-order push toward one branch across runs. This is exactly the behavior
 634 desired from a generic exploration device. A fixed token or fixed soft prompt may still be useful, but
 635 it acts as a persistent prompt prior rather than as an unbiased exploration mechanism.

636 C.3 Direction-Agnostic Coverage

637 The second requirement is directional coverage. Let D be any zero-mean perturbation distribution
 638 in $\mathbb{R}^d$ with covariance Σ_D and variance budget $\text{tr}(\Sigma_D) \leq \rho^2$. For a unit vector b , the first-order
 639 perturbation energy available along b is $\text{Var}_{h \sim D}(b^\top h) = b^\top \Sigma_D b$. This is the quantity that appears
 640 in the local linear logit surrogate used in the main text.

641 **Theorem 3** (Direction-agnostic coverage). *Define*

$$\mathcal{E}(D) := \min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h).$$

642 *Then*

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \leq \frac{\rho^2}{d},$$

643 *with equality if and only if* $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.

644 *Proof.* For any symmetric covariance matrix Σ_D ,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D),$$

645 so $\mathcal{E}(D) = \lambda_{\min}(\Sigma_D)$. Let the eigenvalues of Σ_D be $\lambda_1, \dots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d} \sum_{m=1}^d \lambda_m = \frac{\text{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

646 Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. $\square$

647 This theorem is a statement about second moments, not about Gaussianity by itself. Many distributions
 648 could realize isotropic covariance. We use a Gaussian in RSP because it is the simplest centered,
 649 rotation-symmetric family that matches the scale of pretrained embeddings. The theorem clarifies the
 650 structural advantage: a fixed prompt has $\Sigma_D = 0$ and therefore no across-run exploration, whereas
 651 any anisotropic perturbation leaves at least one branch-sensitive direction less explored than the
 652 isotropic optimum.

653 **Proposition 5** (Covariance pushforward under the value projection). *Let $\bar{h}$ be a centered random*
 654 *vector with covariance $\sigma_E^2 \mathbf{I}_d$, and let $v = W\bar{h}$ for a matrix $W \in \mathbb{R}^{d_v \times d}$. Then*

$$\text{Cov}(v) = \sigma_E^2 W W^\top.$$

655 *Consequently, for every unit vector $u \in \mathbb{S}^{d_v-1}$,*

$$\sigma_E^2 \lambda_{\min}(W W^\top) \leq \text{Var}(u^\top v) = \sigma_E^2 u^\top W W^\top u \leq \sigma_E^2 \lambda_{\max}(W W^\top).$$

656 *In particular, if W is full rank then the projected perturbation has strictly positive variance in every*
 657 *output-space direction.*

658 *Proof.* Since $v = W\bar{h}$ and $\mathbb{E}[\bar{h}] = 0$,

$$\text{Cov}(v) = \mathbb{E}[v v^\top] = W \mathbb{E}[\bar{h} \bar{h}^\top] W^\top = \sigma_E^2 W W^\top.$$

659 For any unit vector u ,

$$\text{Var}(u^\top v) = u^\top \text{Cov}(v) u = \sigma_E^2 u^\top W W^\top u,$$

660 and the Rayleigh quotient bounds give the stated inequalities. $\square$

D Interpretive Connection to Hopfield Retrieval (Optional)

This appendix is included for completeness and is not load-bearing for any quantitative result in the main text. Section 3.3 of the main paper deliberately uses only standard self-attention identities (the exact decomposition of Eq. (2) and the decay bound of Theorem 1); the formal claims there do not assume $\mathbf{K} = \mathbf{V}$, weight-tied layers, or the existence of a global energy. The present appendix records the alternative reading—originating from the Modern Hopfield correspondence [Ramsauer et al., 2021]—under which a single attention update with $\mathbf{K} = \mathbf{V}$ becomes one gradient step on a Hopfield energy. This reading provides additional intuition for why a deterministic forward pass can be exploration-poor and has been a useful lens in the related literature, but it is not used in any proof of Section 3.

We restate the Modern Hopfield correspondence in the form used in this interpretive lens. For a key matrix $\mathbf{K} = (\mathbf{k}_1, \dots, \mathbf{k}_n)$ at a fixed query position, the Hopfield energy is

$$E_{\mathbf{K}}(\sigma) := -\beta^{-1} \log \sum_{j=1}^n \exp(\beta \mathbf{k}_j^\top \sigma) + \frac{1}{2} \|\sigma\|^2.$$

A direct calculation gives the gradient

$$\nabla E_{\mathbf{K}}(\sigma) = \sigma - \mathbf{K}^\top \text{softmax}(\beta \mathbf{K} \sigma).$$

Under (A1) $\mathbf{K} = \mathbf{V}$, the attention update with this $\mathbf{K}$ is therefore $\sigma^{\text{new}} = \sigma - \nabla E_{\mathbf{K}}(\sigma) = \mathbf{K}^\top \text{softmax}(\beta \mathbf{K} \sigma)$, which is one gradient step on $E_{\mathbf{K}}$ at unit step size [Ramsauer et al., 2021].

Lemma 2 (Descent inequality). *If $E_{\mathbf{K}}$ is L_E -smooth on a neighborhood of σ , then the unit-step gradient update $\sigma^{\text{new}} := \sigma - \nabla E_{\mathbf{K}}(\sigma)$ satisfies*

$$E_{\mathbf{K}}(\sigma^{\text{new}}) \leq E_{\mathbf{K}}(\sigma) - (1 - \frac{L_E}{2}) \|\nabla E_{\mathbf{K}}(\sigma)\|^2.$$

Proof. The standard L_E -smooth quadratic upper bound [Nesterov, 2018, Lemma 1.2.3] states $E(\sigma + v) \leq E(\sigma) + \nabla E(\sigma)^\top v + (L_E/2) \|v\|^2$ for any v in the smoothness neighborhood. Applying this to $v = -\nabla E(\sigma)$ gives $E(\sigma - \nabla E) \leq E(\sigma) - \|\nabla E\|^2 + (L_E/2) \|\nabla E\|^2 = E(\sigma) - (1 - L_E/2) \|\nabla E\|^2$. $\square$

Theorem 4 (Deterministic descent and sublevel-set invariance). *Assume (A1) $\mathbf{K} = \mathbf{V}$, (A2) the attention component is weight-tied across blocks, and (A3) $E_{\mathbf{K}}$ is L_E -smooth on a neighborhood of the unperturbed trajectory with $L_E < 2$. Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ is strictly monotone-descending in $E_{\mathbf{K}}$ at every non-stationary point, and the sublevel set $\{\sigma : E_{\mathbf{K}}(\sigma) \leq E_{\mathbf{K}}(\sigma^{(0)})\}$ is forward-invariant.*

Proof. By Lemma 2, $E_{\mathbf{K}}(\sigma^{(l+1)}) \leq E_{\mathbf{K}}(\sigma^{(l)}) - (1 - L_E/2) \|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|^2$. Under (A3), the coefficient $1 - L_E/2 > 0$, so $E_{\mathbf{K}}(\sigma^{(l+1)}) \leq E_{\mathbf{K}}(\sigma^{(l)})$ with strict inequality unless $\nabla E_{\mathbf{K}}(\sigma^{(l)}) = 0$, which proves forward invariance. $\square$

Corollary 1 (Connected-component containment for the unit-step iterate). *Let $S := \{\sigma : E_{\mathbf{K}}(\sigma) \leq E_{\mathbf{K}}(\sigma^{(0)})\}$ and let $C(\sigma^{(0)})$ denote the connected component of S containing $\sigma^{(0)}$. Define*

$$\text{sep}(C(\sigma^{(0)})) := \inf\{\|x - y\| : x \in C(\sigma^{(0)}), y \in S \setminus C(\sigma^{(0)})\},$$

with the convention $\text{sep} = +\infty$ if S is connected. Assume $\text{sep}(C(\sigma^{(0)})) > 0$. Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ remains in $C(\sigma^{(0)})$ for all $l \geq 0$ whenever

$$\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\| < \text{sep}(C(\sigma^{(0)})) \quad \text{for every } l.$$

Proof. By Theorem 4, every iterate lies in S . Suppose by induction $\sigma^{(l)} \in C(\sigma^{(0)})$. The displacement $\sigma^{(l+1)} - \sigma^{(l)} = -\nabla E_{\mathbf{K}}(\sigma^{(l)})$ has Euclidean length $\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|$. A jump to a distinct component $C' \subseteq S$ would require this length to be at least $\inf\{\|x - y\| : x \in C(\sigma^{(0)}), y \in C'\} \geq \text{sep}(C(\sigma^{(0)}))$, which the hypothesis excludes. Hence $\sigma^{(l+1)} \in C(\sigma^{(0)})$. $\square$

698 **Positivity of sep.** $\text{sep}(C) > 0$ holds whenever the components of S are spatially separated by a
699 region of strictly higher energy than $E_{\mathbf{K}}(\sigma^{(0)})$, which we treat as a problem-dependent assumption
700 rather than a structural property: it requires that the relevant components of the Hopfield energy do
701 not share closure boundary points at level $E_{\mathbf{K}}(\sigma^{(0)})$. If $\text{sep} = 0$ the corollary becomes vacuous; in
702 that case the trajectory is still in S by Theorem 4, but is not formally trapped in $C(\sigma^{(0)})$.

703 **Caveats and scope.** Theorem 4 depends on three assumptions that are individually mild but jointly
704 idealized for real transformers. (A1) $\mathbf{K} = \mathbf{V}$ collapses the key and value projections; this is the
705 standard Modern Hopfield assumption [Ramsauer et al., 2021] and isolates the energy-iteration aspect
706 of attention from the additional value-projection degree of freedom. (A2) weight-tying treats the
707 attention component as a single repeated map; without it, each block has its own energy and the
708 trajectory does not minimize a single fixed function across blocks. (A3) is a smoothness condition on
709 $E_{\mathbf{K}}$. For the Hopfield energy used here, $\nabla^2 E_{\mathbf{K}}(\sigma) = \mathbf{I} - \beta \mathbf{K}^\top (\text{diag}(p_\sigma) - p_\sigma p_\sigma^\top) \mathbf{K}$ where $p_\sigma =$
710 $\text{softmax}(\beta \mathbf{K} \sigma)$, so the operator-norm bound on $\nabla^2 E_{\mathbf{K}}$ depends on the inverse-temperature β and
711 on $\|\mathbf{K}\|$. (A3) holds whenever $\beta \|\mathbf{K}\|^2$ is sufficiently small, which is the standard regime in which the
712 Hopfield iterate converges in a small number of steps. The theorem therefore characterizes an idealized
713 tendency of attention-only iteration in this regime; it is not a literal property of every transformer
714 block. In particular, sublevel-set confinement follows from Theorem 4, whereas connected-component
715 confinement additionally requires the separation and per-step gradient assumptions of Corollary 1. Its
716 role in the main text is to supply an idealized source of exploration deficit that an injected stochastic
717 perturbation addresses; the empirical link is the cross-layer attention decay observed in Figure 1 and
718 the trajectory diversity gain reported in Section 4.4.

E Proofs for the Transformer-Level Mechanism

This appendix collects the formal proofs underlying Sections 3.3–3.4 of the main text.

E.1 Exact Attention Decomposition and Magnitude Control

Proposition 6 (Attention decomposition). *Let $\alpha_{ij}^{(l)}$ be the attention weights for query position i at layer l , with n real tokens and k random tokens. Define*

$$w_{r,i}^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)}$$

and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(l)} := \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}}.$$

Then

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)},$$

where

$$\tilde{\mathbf{x}}_i^{(l+1)} := \sum_{j=1}^n \tilde{\alpha}_{ij}^{(l)} \mathbf{v}_j^{(l)} \quad \text{and} \quad \boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

Proof. The attention output is

$$\mathbf{x}_i^{(l+1)} = \sum_{j=1}^n \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} + \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

By definition, $\sum_{j=1}^n \alpha_{ij}^{(l)} = 1 - w_{r,i}^{(l)}$, so

$$\sum_{j=1}^n \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)}) \sum_{j=1}^n \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)}.$$

Substituting this identity gives the decomposition. □

Corollary 2 (Magnitude scales with random-token attention). *For every realization of the random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

Proof. By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(l)}\| = \left\| \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)} \right\| \leq \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \|\mathbf{v}_{r_j}^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

□

Corollary 2 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(l)}$ is small, the random contribution must be small as well.

737 E.2 Attention Decay Bound

738 **Theorem 5** (Attention decay). *The total attention on random tokens satisfies*

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)} / \sqrt{d})}, \quad (7)$$

739 where $\Delta_i^{(l)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$, is the real-vs-random logit gap.

740 *Proof.* Let $s_j = \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d}$ for real keys and $s_{r_{j'}} = \mathbf{q}_i^\top \mathbf{k}_{r_{j'}} / \sqrt{d}$ for random keys. By definition
 741 of $\Delta_i^{(l)}$, each real key satisfies $s_j \geq \Delta_i^{(l)} / \sqrt{d} + s_{r,\max}$ where $s_{r,\max} = \max_{j'} s_{r_{j'}}$. Therefore
 742 $\exp(s_j) \geq \exp(\Delta_i^{(l)} / \sqrt{d}) \cdot \exp(s_{r,\max})$ for every real key. Summing over n real keys gives

$$\sum_{j=1}^n \exp(s_j) \geq n \cdot \exp(\Delta_i^{(l)} / \sqrt{d}) \cdot \exp(s_{r,\max}) \geq n \cdot \exp(\Delta_i^{(l)} / \sqrt{d}) \cdot \frac{1}{k} \sum_{j'=1}^k \exp(s_{r_{j'}}).$$

743 Let $R := \sum_{j'=1}^k \exp(s_{r_{j'}})$. Then

$$w_{r,i}^{(l)} = \frac{R}{\sum_{j=1}^n \exp(s_j) + R} \leq \frac{R}{n \cdot \exp(\Delta_i^{(l)} / \sqrt{d}) \cdot R/k + R} = \frac{k}{k + n \cdot \exp(\Delta_i^{(l)} / \sqrt{d})}.$$

744 □

745 **Corollary 3** (Sequence-wise annealing under a fixed logit gap). *For fixed k and $\Delta_i^{(l)}$, the upper*
 746 *bound*

$$f(n) = \frac{k}{k + n \exp(\Delta_i^{(l)} / \sqrt{d})}$$

747 *is strictly decreasing in n . Consequently, cache growth alone decreases the maximum perturbation*
 748 *mass compatible with that gap.*

749 *Proof.* Differentiate $f(n)$ with respect to n :

$$f'(n) = -\frac{k \exp(\Delta_i^{(l)} / \sqrt{d})}{\left(k + n \exp(\Delta_i^{(l)} / \sqrt{d})\right)^2} < 0.$$

750 Thus $f(n)$ is strictly decreasing. □

751 **Corollary 4** (Vanishing random contribution under annealing). *If $n \rightarrow \infty$ along decoding while k*
 752 *and $\Delta_i^{(l)}$ remain fixed, then $w_{r,i}^{(l)} \rightarrow 0$. Moreover, for any fixed realization of the k random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\| \rightarrow 0.$$

753 *Since the implementation samples finite-dimensional vectors and k is finite, the displayed maximum*
 754 *is almost surely finite.*

755 *Proof.* Corollary 3 implies $w_{r,i}^{(l)} \rightarrow 0$. The bound on $\|\boldsymbol{\eta}_i^{(l)}\|$ is Corollary 2. Since the multiplier
 756 $\max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|$ is finite for any fixed realization of finitely many sampled values, the product
 757 converges to zero. □

758 E.3 Feed-Forward Propagation under Local Lipschitz Control

759 A transformer block applies an attention sublayer followed by a feed-forward sublayer with residual
 760 connections:

$$\mathbf{x}_i^{(l,\text{attn})} = \mathbf{x}_i^{(l)} + \text{Attn}^{(l)}(\mathbf{x}_i^{(l)}), \quad \mathbf{x}_i^{(l+1)} = \mathbf{x}_i^{(l,\text{attn})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn})}).$$

761 Let $\mathbf{x}_i^{(l,\text{attn},\text{RSP})}$ and $\mathbf{x}_i^{(l,\text{attn},\text{base})}$ denote the post-attention states under RSP and the unperturbed forward
 762 pass at the same input $\mathbf{x}_i^{(l)}$.

763 **Lemma 3** (Per-block FFN propagation). *The post-attention difference satisfies*

$$\|\mathbf{x}_i^{(l, \text{attn}, \text{RSP})} - \mathbf{x}_i^{(l, \text{attn}, \text{base})}\| \leq \|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|.$$

764 *If, in addition, $F^{(l)}$ is $L^{(l)}$ -Lipschitz on a bounded set containing both $\mathbf{x}_i^{(l, \text{attn}, \text{RSP})}$ and $\mathbf{x}_i^{(l, \text{attn}, \text{base})}$,*
 765 *then*

$$\|\mathbf{x}_i^{(l+1, \text{RSP})} - \mathbf{x}_i^{(l+1, \text{base})}\| \leq (1 + L^{(l)}) (\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|).$$

766 *Proof.* The unperturbed attention sublayer produces $\mathbf{x}_i^{(l, \text{attn}, \text{base})} = \mathbf{x}_i^{(l)} + \tilde{\mathbf{x}}_i^{(l+1)}$, while the RSP
 767 attention sublayer produces $\mathbf{x}_i^{(l, \text{attn}, \text{RSP})} = \mathbf{x}_i^{(l)} + (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}$ by Proposition 6. Subtracting,

$$\mathbf{x}_i^{(l, \text{attn}, \text{RSP})} - \mathbf{x}_i^{(l, \text{attn}, \text{base})} = \boldsymbol{\eta}_i^{(l)} - w_{r,i}^{(l)} \tilde{\mathbf{x}}_i^{(l+1)},$$

768 and the triangle inequality gives the first bound. For the second bound, write the post-FFN difference
 769 as

$$[\mathbf{x}_i^{(l, \text{attn}, \text{RSP})} + F^{(l)}(\mathbf{x}_i^{(l, \text{attn}, \text{RSP})})] - [\mathbf{x}_i^{(l, \text{attn}, \text{base})} + F^{(l)}(\mathbf{x}_i^{(l, \text{attn}, \text{base})})].$$

770 By Lipschitz continuity, $\|F^{(l)}(\mathbf{x}_i^{(l, \text{attn}, \text{RSP})}) - F^{(l)}(\mathbf{x}_i^{(l, \text{attn}, \text{base})})\| \leq L^{(l)} \|\mathbf{x}_i^{(l, \text{attn}, \text{RSP})} - \mathbf{x}_i^{(l, \text{attn}, \text{base})}\|$.
 771 The triangle inequality combined with the first bound gives the second. $\square$

772 **Corollary 5** (Cross-block propagation control). *Define $\Delta^{(l)} := \mathbf{x}_i^{(l, \text{RSP})} - \mathbf{x}_i^{(l, \text{base})}$ with $\Delta^{(0)} = \mathbf{0}$*
 773 *(matched inputs at layer 0), and let $L_{\text{block}}^{(l)}$ denote a common local Lipschitz constant valid for both the*
 774 *baseline block map $B_{\text{base}}^{(l)} : \mathbf{x} \mapsto \mathbf{x} + \text{Attn}^{(l)}(\mathbf{x}) + F^{(l)}(\mathbf{x} + \text{Attn}^{(l)}(\mathbf{x}))$ and the RSP-augmented*
 775 *block map $B_{\text{RSP}}^{(l)}$ (which replaces $\text{Attn}^{(l)}$ with the attention computation that includes the random*
 776 *tokens) on a ball containing both trajectories. Such a constant exists for standard transformer blocks*
 777 *because Attn , LayerNorm , and the FFN sublayer are each locally Lipschitz on any bounded set*
 778 *[Xiong et al., 2020, Kim et al., 2021], and the random-key augmentation only adds a finite number of*
 779 *additional softmax inputs at every layer. Then*

$$\|\Delta^{(l+1)}\| \leq (1 + L_{\text{block}}^{(l)}) \|\Delta^{(l)}\| + (1 + L^{(l)}) (\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|).$$

780 *Iterating across L blocks with $\Delta^{(0)} = \mathbf{0}$ yields*

$$\|\Delta^{(L+1)}\| \leq \sum_{l=0}^L \prod_{l'=l+1}^L (1 + L_{\text{block}}^{(l')}) \cdot (1 + L^{(l)}) (\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|).$$

781 *Proof.* Decompose the cross-block discrepancy at layer $l + 1$ into two contributions:

$$\Delta^{(l+1)} = \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l, \text{RSP})}) - B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l, \text{base})})}_{\text{(I) propagation of prior } \Delta^{(l)}} + \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l, \text{base})}) - B_{\text{base}}^{(l)}(\mathbf{x}^{(l, \text{base})})}_{\text{(II) RSP-induced perturbation injected at block } l},$$

782 where $B_{\text{RSP}}^{(l)}$ replaces $\text{Attn}^{(l)}$ with the RSP-augmented attention. Term (I) is bounded via the block
 783 Lipschitz constant: $\|(I)\| \leq L_{\text{block}}^{(l)} \|\Delta^{(l)}\|$. With residuals included one gets the factor $(1 + L_{\text{block}}^{(l)})$
 784 when also propagating the residual term $\Delta^{(l)}$ itself. Term (II) is exactly the per-block injection from
 785 Lemma 3, bounded by $(1 + L^{(l)}) (\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|)$. Combining via the triangle inequality
 786 gives the recursion. The summed bound follows by unrolling the recursion: with $\Delta^{(0)} = \mathbf{0}$, induction
 787 on L shows that $\|\Delta^{(L+1)}\|$ is at most the displayed sum. $\square$

788 **Boundedness of $L^{(l)}$ for standard architectures.** Modern transformer blocks combine LayerNorm,
 789 GELU or SwiGLU activations, and linear projections. LayerNorm has Lipschitz constant on the
 790 order of $\|\gamma\| / \inf \|x\|$ on every set bounded away from the origin, where γ is the LayerNorm scale
 791 parameter [Xiong et al., 2020]. GELU and SwiGLU are smooth and therefore locally Lipschitz on
 792 bounded activation regions, and linear projections have Lipschitz constant equal to the spectral norm
 793 of the weight matrix. The composed self-attention sublayer has a Lipschitz constant that depends
 794 on the spectral norms of $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$ and the inverse-temperature β as analyzed by Kim et al.
 795 [2021]. On the bounded activation regions traversed during a single forward pass, the per-block
 796 Lipschitz constants in Corollary 5 are therefore finite. For a fixed model depth this yields a finite
 797 cumulative bound on $\|\Delta^{(L+1)}\|$; stronger depth-uniform non-expansion would require additional
 798 contraction assumptions not imposed here. This is the formal sense in which the injected perturbation
 799 stays controlled across a finite forward pass, while the size of each new injection is set by $w_{r,i}^{(l)}$.

800 E.4 Proofs for Output Distribution Shift

801 The main text §3.4 reports the surrogate-level top- K entry result (Proposition 2) and the multi-
 802 path Pass@ N accumulation (Proposition 3). For completeness, we also include a pairwise rank
 803 randomness statement below.

804 E.4.1 Autoregressive Formulation

805 In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^T p(y_t \mid x, y_{<t}),$$

806 where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by
 807 RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single
 808 global distribution.

809 E.4.2 Local Linearization of Logits under RSP

810 Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector $\bar{h}_l$ from the RSP definition
 811 (Eq. (1) in §3.1), treated as a per-token surrogate. The actual implementation uses a length- L prompt
 812 $\mathbf{H} = (h_1, \dots, h_L)$ with L independent draws; the propositions below identify how each individual
 813 $\bar{h}_l$ shifts the logits, and the multi-token compounding through softmax-weighted attention is reported
 814 empirically in Figure 1 and Section 4.4. We model the perturbed logits as a function $z_i(\bar{h})$, assuming
 815 local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

816 where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

817 This approximation is valid for sufficiently small $\|\bar{h}\|$ and is used as a first-order surrogate for the
 818 direction of perturbation rather than as an exact global model of the transformer logits.

819 For the two propositions below we isolate the only support property that is actually used. Let D
 820 denote the centered prompt law. We assume

$$(S1) \quad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

821 The Gaussian instantiation in Section 3.4 satisfies (S1), but the proofs below require only this support
 822 condition.

823 E.4.3 Proposition (Stochastic Ranking) and proof

824 **Proposition 7** (Stochastic ranking). *If there exists a token pair (i, j) such that $a_i \neq a_j$ and the*
 825 *centered prompt law satisfies (S1), then $0 < \Pr_{\bar{h}}(z_i^{\text{lin}}(\bar{h}) > z_j^{\text{lin}}(\bar{h})) < 1$.*

826 *Proof.* Under the local affine surrogate,

$$z_i^{\text{lin}}(\bar{h}) - z_j^{\text{lin}}(\bar{h}) = (z_i - z_j) + (a_i - a_j)^\top \bar{h}.$$

827 Let $b_{ij} := a_i - a_j \neq 0$. The perturbed logit difference

$$\Delta_{ij}(\bar{h}) := (z_i - z_j) + b_{ij}^\top \bar{h}$$

828 vanishes on the affine hyperplane

$$H_0 := \{\bar{h} \in \mathbb{R}^d : (z_i - z_j) + b_{ij}^\top \bar{h} = 0\}.$$

829 Because $b_{ij} \neq 0$, H_0 has codimension one and partitions $\mathbb{R}^d$ into the two nonempty open half-spaces

$$H_+ := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} > 0\}, \quad H_- := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} < 0\}.$$

830 By (S1), both H_+ and H_- have positive probability, which yields

$$0 < \Pr_{\bar{h}}(\Delta_{ij}(\bar{h}) > 0) < 1.$$

831

□

832 **E.4.4 Proof of Proposition 2 (Top-K Entry)**

833 Let $J_K(s_t)$ be the baseline top- K set at state s_t , and let $i \notin J_K(s_t)$. Define

$$\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K-1\}.$$

834 If $\mathcal{G}_{i,K}$ contains a nonempty open set and the centered prompt law satisfies (S1), then

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0.$$

835 *Proof.* Under the local affine surrogate, token i belongs to the surrogate top- K set if and only if at
 836 most $K-1$ tokens outrank it, i.e., if and only if $\bar{h} \in \mathcal{G}_{i,K}$. For each subset S of token indices with
 837 $i \notin S$ and $|S| \leq K-1$, define

$$\mathcal{O}_S := \{\bar{h} : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h}) \text{ for all } j \in S, z_i^{\text{lin}}(\bar{h}) > z_j^{\text{lin}}(\bar{h}) \text{ for all } j \notin S \cup \{i\}\}.$$

838 Each $\mathcal{O}_S$ is an intersection of open half-spaces and is therefore an open subset of $\mathbb{R}^d$. By construction,
 839 every $\bar{h} \in \mathcal{O}_S$ has token i ranked $|S|+1 \leq K$ in the affine surrogate, so $\mathcal{O}_S \subseteq \mathcal{G}_{i,K}$. Hence

$$\bigcup_{S: i \notin S, |S| \leq K-1} \mathcal{O}_S \subseteq \mathcal{G}_{i,K}.$$

840 The reverse inclusion fails only on the measure-zero set of $\bar{h}$ that produce ties, which we do not need:
 841 the assumption that $\mathcal{G}_{i,K}$ contains a nonempty open set is sufficient. Condition (S1) therefore implies

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0,$$

842 which is exactly the event that token i enters the surrogate top- K set. $\square$

843 **E.4.5 Sample complexity corollary and proof**

844 **Corollary 6** (Sample complexity). Let $\mu_i = \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$. Within N independent RSP
 845 samples $\bar{h}_1, \dots, \bar{h}_N$, the probability of at least one admission is at least $1 - (1 - \mu_i)^N \geq 1 -$
 846 $\exp(-N\mu_i)$.

847 *Proof.* The samples $\{\bar{h}_n\}_{n=1}^N$ are independent, so

$$\Pr(\text{no admission within } N \text{ samples}) = \prod_{n=1}^N \Pr(\bar{h}_n \notin \mathcal{G}_{i,K}) = (1 - \mu_i)^N.$$

848 The complement gives the first inequality. The second inequality is standard: $1 - \mu_i \leq \exp(-\mu_i)$,
 849 hence $(1 - \mu_i)^N \leq \exp(-N\mu_i)$. $\square$

850 **E.4.6 Proof of Proposition 3 (Exploration-meets-multi-path Pass@ N gain)**

851 Fix a problem x and let $p_n(x) \geq p_{\min}(x) > 0$ be the per-run success probability of the n -th RSP run
 852 as in (M1). Under (M1)+(M2),

$$\Pr(\text{Pass@}(N+1) \text{ correct on } x) \geq \max\{p_{\text{base}}(x), 1 - (1 - p_{\min}(x))^N\} \geq 1 - e^{-Np_{\min}(x)}.$$

853 *Proof.* Let A_0 be the event that the deterministic baseline run reaches a trajectory in $\mathcal{T}^*(x)$, and
 854 let A_n ($n = 1, \dots, N$) be the event that the n -th RSP run reaches such a trajectory. Pass@ $(N+1)$
 855 succeeds on x iff at least one of $A_0, A_1, \dots, A_N$ occurs.

856 By assumption $p_{\text{base}}(x) := \Pr(A_0) \in \{0, 1\}$ and $\Pr(A_n) = p_n(x) \geq p_{\min}(x)$ for each $n \geq 1$.
 857 Independence of the N RSP runs (M2) gives

$$\Pr\left(\bigcap_{n=1}^N A_n^c\right) = \prod_{n=1}^N (1 - p_n(x)) \leq \prod_{n=1}^N (1 - p_{\min}(x)) = (1 - p_{\min}(x))^N,$$

858 using $p_n(x) \geq p_{\min}(x)$ in the inequality. Therefore

$$\Pr\left(\bigcup_{n=1}^N A_n\right) = 1 - \Pr\left(\bigcap_{n=1}^N A_n^c\right) \geq 1 - (1 - p_{\min}(x))^N.$$

859 The $\text{Pass}@ (N+1)$ event $A_0 \cup A_1 \cup \dots \cup A_N$ has probability at least $\max\{\Pr(A_0), \Pr(\bigcup_{n \geq 1} A_n)\}$,
 860 giving the first claimed inequality. The second inequality follows from the standard bound $1 - x \leq e^{-x}$
 861 for $x \in [0, 1]$ applied with $x = p_{\min}(x)$:

$$(1 - p_{\min}(x))^N \leq e^{-Np_{\min}(x)},$$

862 hence $1 - (1 - p_{\min}(x))^N \geq 1 - e^{-Np_{\min}(x)}$. □

863 **E.4.7 Trajectory-Level Compounding**

864 Since autoregressive generation composes conditional distributions across steps, perturbations at
 865 early steps can lead to distinct trajectories. Let $\mathcal{R}_{k,K}(x)$ denote the set of trajectories containing
 866 at least one token outside the baseline top- K set. If RSP increases the accessibility of such tokens
 867 at early steps, then the probability of sampling trajectories in $\mathcal{R}_{k,K}(x)$ increases under repeated
 868 sampling. The key point is temporal compounding: one early top- K entry changes the next decoding
 869 state, which changes the next candidate set, and so on. This is the heuristic bridge from local rank
 870 perturbations to global trajectory diversity that motivates assumption (M1) of Proposition 3.

871 F Attention Mass Analysis

872 This appendix formalizes the per-token attention mass reported in Figure 1 and the exclusion criteria
 873 applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh
 874 RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both
 875 problem variability and RSP-vector variability.

876 F.1 Per-Token Attention Mass

877 For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where
 878 the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$
 879 denote the post-softmax attention weight at layer ℓ , head h , from query position i to key position j ,
 880 satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and
 881 RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention
 882 mass that query i directs to each group at layer ℓ is

$$\text{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^H \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \quad \text{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^H \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \quad (8)$$

883 Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a
 884 single question (resp. RSP) token receives on average under the same softmax denominator. Question
 885 tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both
 886 groups.

887 F.2 Heatmap Aggregation

888 We partition the layer indices and the reasoning position indices each into five equal-size bins
 889 $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_{\bullet}^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \text{PTM}_{\bullet}[\ell, i]. \quad (9)$$

890 Figure 1 reports the sample average of Eq. (9) over the 500 valid samples.

891 F.3 Excluding the `\boxed{\dots}` Span

892 The reasoning position range terminates strictly before the final `\boxed{\dots}` span. Chain-of-
 893 thought outputs decompose into a *text* (content) component and a *pattern* (template) component
 894 that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The
 895 `\boxed{\dots}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the
 896 answer format rather than to extend reasoning. Including it would therefore inject a template-driven
 897 attention signature unrelated to reasoning dynamics.

898 F.4 Excluding the Final Two Layers

899 The layer axis further excludes the last two transformer layers. Two well-documented properties of
 900 late layers, none specific to our setup, motivate this choice.

901 **Output-projection specialization.** Late-layer hidden states lie in an approximately linear rela-
 902 tionship with vocabulary logits and therefore function as a progressive linear readout rather than
 903 as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward
 904 modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021].
 905 Attention in these layers therefore reflects output formation rather than reasoning computation.

906 **Register-token hypothesis.** Uninformative patch tokens in Vision Transformers have been shown
 907 to be repurposed as *registers*: storage slots that absorb attention in deeper layers [Darcet et al., 2024].
 908 Because RSPs share a similar profile as continuous tokens without pretrained semantic content, a

similar repurposing in late transformer layers cannot be ruled out. Excluding the final two layers removes this potential register-induced confound from the reasoning-phase signal.

The two lines of evidence above identify known late-layer phenomena — linear readout and potential register repurposing — whose attention signatures are orthogonal to reasoning-phase dynamics. Removing the final two layers preempts their contamination of the reasoning-phase signal, yielding a principled, literature-grounded exclusion rather than a model-specific tuning decision.

F.5 Additional Suffix Heatmaps

Figure 5 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows the layer-wise and sequence-wise decay predicted by Theorem 1.

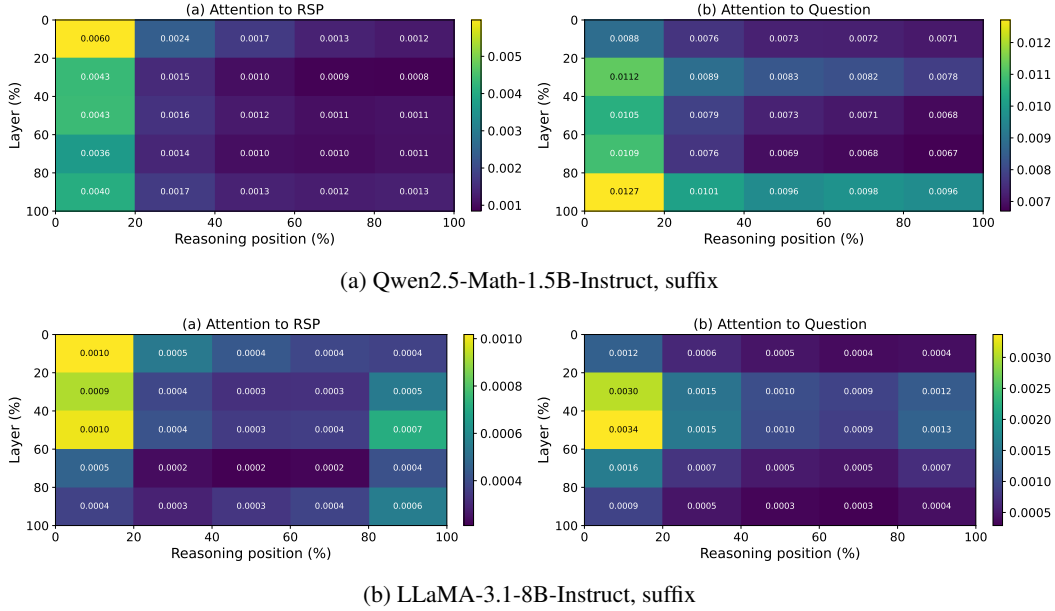


Figure 5: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 1: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

921 **G GRPO Formulation**

922 For a given prompt q , the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G responses $\{o_i\}_{i=1}^G$ with
 923 corresponding rewards $\{R_i\}_{i=1}^G$. The advantage of each response is computed by normalizing the
 924 group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \quad (10)$$

925 The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t} \right) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right], \quad (11)$$

926 where $r_{i,t} = \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, ε is the clipping
 927 range, and β controls the strength of the KL penalty against the reference policy π_{ref} .

928 H Broader Impacts

929 This work provides empirical and theoretical analysis of how random embedding injection affects
930 the reasoning behavior of large language models. The contributions are primarily foundational,
931 with potential downstream implications for reinforcement learning-based post-training of reasoning
932 models such as GRPO.

933 **Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-
934 based LLM post-training by providing richer learning signals within group-relative advantage estima-
935 tion. This has the potential to reduce compute costs associated with alignment and reasoning-oriented
936 fine-tuning, making such methods more accessible to research groups with limited resources.

937 **Potential concerns.** Methods that expand the effective output support of LLMs may increase the
938 reachability of low-probability tokens, which includes both desirable alternative reasoning paths
939 and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems
940 should combine it with existing safety filtering and alignment mechanisms rather than treating it as a
941 standalone diversity source.

942 **Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new
943 model or dataset, the direct societal impact is limited. The broader implications described above are
944 contingent on future work integrating RSP into applied training pipelines.

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